KRISTIYAN SAKALYAN

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Page ◇ Scholar ◇ Github ◇ LinkedIn

EDUCATION

Technical University of Munich (TUM) *September 2025 (expected)*
M.Sc. in Data Engineering & Analytics
Expected average grade: 1.2
Current rank: Top 4%

Technical University of Munich (TUM) *October 2022*
B.Sc. in Computer Science
Thesis title: "Multi-object tracking for annotating markers for a motion capture system"
Thesis grade: 1.3

RESEARCH INTERESTS

- AI4Science
- Generative AI
- Graphs and Geometric Deep Learning

RESEARCH EXPERIENCE

Modeling Microenvironment Trajectories on Spatial Transcriptomics Jan 2024 – Present
Supervisors: F. Guerranti, A. Palma, F. Theis, S. Günnemann TUM

- Investigating graph- and point cloud-based methodologies to model single-cell dynamics, leveraging Flow Matching and Optimal Transport for time-resolved spatial data.

Conditional Flow Matching for Temporal Graphs Oct 2024 – Jan 2024
Supervisors: F. Guerranti, A. Palma, S. Günnemann TUM

- Extend previous works by applying flow matching to dynamic temporal graphs. [Code]

Continual Learning for Domain Incremental Semantic Segmentation Apr 2024 – Oct 2024
Supervisors: G. Ghazaei, R. Avecado, M. Fornasier ZEISS & MDSI & TUM

- Developed a new method to reduce catastrophic forgetting in domain-incremental semantic segmentation using contrastive learning [Report]

Explainable Document Embeddings Oct 2023 – Apr 2024
Supervisors: G. Hagerer, G. Groh TUM

- Enhanced the Bag-of-Concepts framework with an automated approach to concept labeling for document embeddings using LLMs [Code]

Patch by Patch Coarse-to-Fine 3D Shape Refinement Oct 2023 – Feb 2024
Supervisors: L. Li, M. Nießner TUM

- Developed a method to enhance geometric detail in 3D shape generation through patch-based coarse-to-fine refinement of sparse point clouds [Report, Poster]

Multi-object tracking for annotating markers for a motion capture system — Nov 2021 – Mar 2022

Supervisors: S. Banik, A.C. Knoll — TUM & reFit

- Conducted research on automating marker annotation for custom skeletons, integrating AI and motion capture systems as part of my Bachelor's thesis at TUM and reFit Systems GmbH.

ACHIEVEMENTS

Deutschlandstipendium — *Oct 2023*

Awarded by TUM and Infineon Technologies AG for outstanding academic achievements. [PDF]

WORK EXPERIENCE

Machine Learning Engineer — Jan 2025 - Present

Working Student — Rohde & Schwarz

- Working on the computer vision method used in QPS Walk 2000.

Data Engineer & Scientist — Mar 2023 - Jan 2025

Working Student — Airbus Defence & Space

- Conducted machine learning research on 4D flight trajectory prediction, implementing transformer-based, LSTM, Neural ODE, LSTM-ODE hybrid, Conditional Flow Matching, and Difusion models.
- Engineered data pipelines with Dask for large-scale processing and redesigned real-time prediction services for airplane ETA using Python, Docker, Kubernetes, Helm, TerraForm, Azure Pipelines, Grafana, Prometheus.

Software Engineer — Aug 2022 - Feb 2023

Working Student — Techpilot

- Led sprint planning, code reviews, and iterative delivery in a collaborative agile environment.
- Enhanced the web platform by building features with React, TypeScript, and Material-UI, connecting buyers with suppliers.

Junior Software Engineer — Oct 2020 - Jul 2022

Working Student — Techpilot

- Maintained and expanded client-server applications, applying React, TypeScript, and Spring Framework for scalable development.
- Contributed to front-end and back-end enhancements, optimizing user experience and system efficiency.

R&D Engineer — Dec 2019 - May 2020

Working Student — HARMAN International

- Developed a POI aggregation web service prototype, consolidating and visualizing location data from various sources on a map.
- Designed algorithms to eliminate duplicates, ensuring comprehensive, non-redundant data aggregation.

SKILLS/HOBBIES

Programming Languages	Python, C++, SQL, Java, JavaScript, TypeScript
Machine Learning Tools	Pytorch, Sklearn, Pandas, Polars, Numpy, TensorBoard, W&B Hydra, Slurm PySpark, Dask
Cloud & Infrastructure	Azure, Azure Pipelines, Terraform, Kubernetes, Docker, Helm
Monitoring Tools	Grafana (Loki, Tempo), Prometheus
Databases	PostgreSQL, SQLite, MongoDB, Redis, RisingWave
Web	React.js, Material-UI, Spring, Express, Node.JS
Hobbies	hiking, running, cycling, swimming, handcrafting

EXTRACURRICULAR TRAINING

Natural Language Processing Specialization Oct 2023

- Certificate

Geometric Deep Learning Oct 2024

- Lecture and Exercises

LANGUAGES

Bulgarian	Native	**German**	C1
English	Proficient	**Spanish**	Beginner